

جامعة السلطان قابوس
Sultan Qaboos University

Sultan Qaboos University College of Science

Department of Computer Science

COMP4701 - Web Application Development

StudySpace: Campus Quiet Study Spot Finder

Project Part 1

Submitted to: Dr. Abdullah Al-Hamdani

Members:

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Date: 27/10/2026

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1. Project Information

Objectives:

1. Real-Time Spot Discovery:
   Enable students to locate, filter, and view live availability of quiet study spots across campus based on noise level, seating capacity, power outlet access and equipment.

2. Crowdsourced and Admin Updates:
   Provide an easy interface for students to report current noise levels/occupancy while allowing campus administrators to manage facility details, hours and maintenance notices.

3. Interactive Reservation and Booking:
   Allow students to reserve study rooms or individual quiet spots in advance to optimize study time during peak exam periods.

Motivation and Real-World Relevance:

Finding a suitable and a quiet study location on a university campus is a persistent challenge for students. Libraries often reach maximum capacity during peak hours, while alternative locations remain underutilized simply because students do not know they exist or whether they have available seats and power outlets.

This inefficiency leads to wasted time, increased stress and sub-optimal environments. StudySpace addresses this issue by aggregating campus study locations into single, centralized web application.

Benefits And Specific Problems Solved:

- Eliminates Campus Wandering:
  Saves time by showing real-time occupancy and noise levels before a student walks across campus.
- Optimizes Campus Resource:
  Distributes student density evenly across primary and secondary study spaces on campus.
- Filterable Comfort Criteria:
  Helps students find spots tailored to specific study needs (e.g: quiet zone vs group study, proximity to power outlets, desktop monitors, whiteboards).

Potential Users:

1. Students (Primary Users):
   Need to find, filter, rate and reserve quiet study locations quickly.

2. Campus Facility / Library Administrators: Need to manage study spaces, set operating hours, review crowdsourced reports, track utilization trends, and post maintenance announcements.

2. User Interviews

Two user interviews were conducted to define key functional and operational requirements for the system.

Interview 1: CS Student (End User)

- Role: Undergraduate Student
- Key Pain Points:
  Expressed frustration over walking to main library during midterms only to find every seat taken. Highlighted the importance of knowing whether power outlets and Wi-Fi quality are good before arriving.
- Features Requests:
  Real-time noise/crowd level indicators, power outlet availability filters, and a quick booking feature for private group rooms.

Interview 2: University Library Assistant (Administrator)

- Role: Campus Administrator
- Key Pain Point:
  Difficulties managing room reservations manually and handling complaints regarding noisy group areas adjacent to quiet zones.
- Feature Requests:
  An admin dashboard to flag or update room statuses, set noise policies per room, monitor occupancy reports, and moderate incorrect crowdsourced user tags.

Impact on Project Requirements:

1. From Student Feedback:
   Added filtering by Power Outlets and Crowdsourced Noise Level Reporting System (Quiet, Moderate, Loud).

2. From Admin Feedback: Incorporated an Admin Management Dashboard for updating spot availability, moderating user feedback, and managing room bookings.

3. Project Requirements

Functional Requirements:

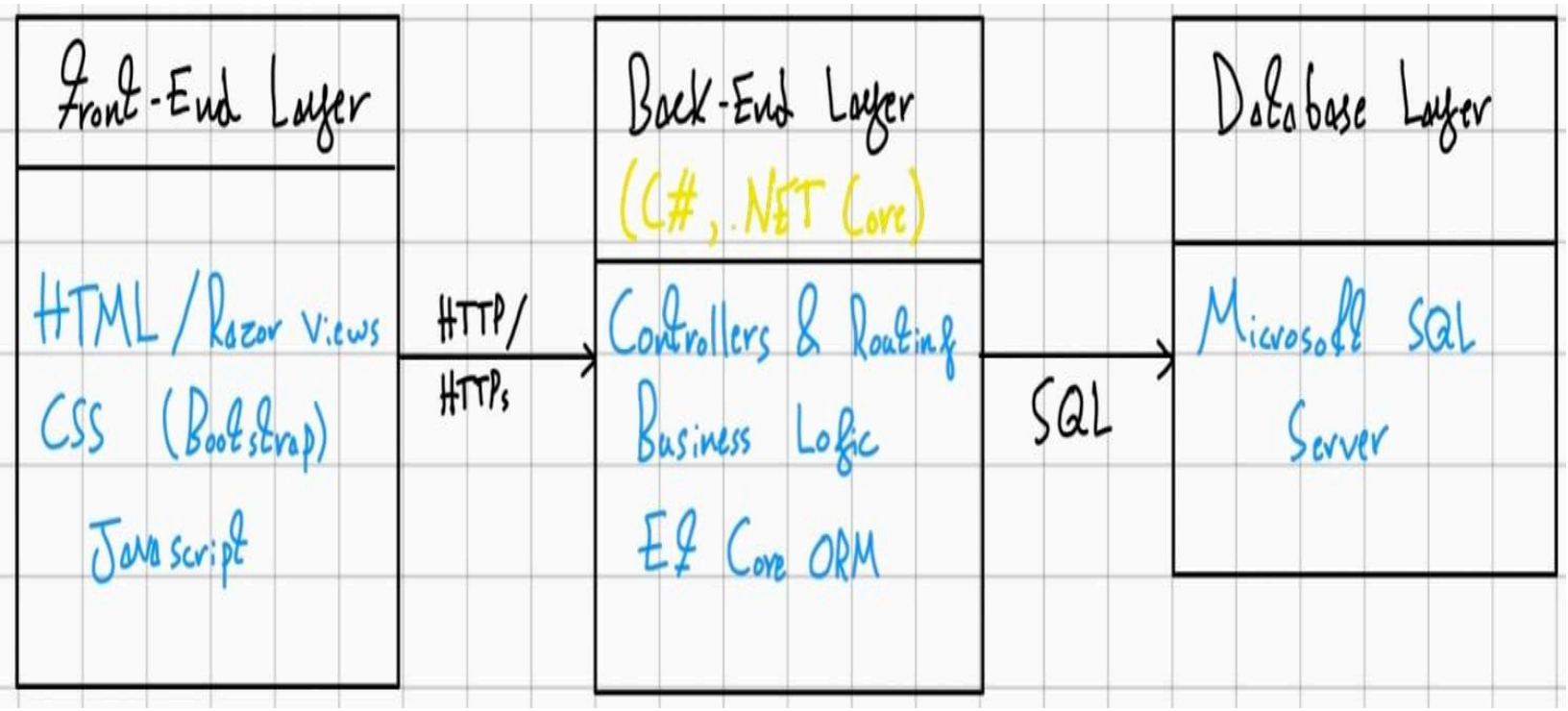
1. User Authentication and Profiles:
   Users should be able to register, log in and manage their profile using university credentials.
2. Spot Search and Filtering:
   Users shall be able to search study spots by building, noise level (Quiet, Moderate, Loud), power outlet availability, and seat count.
3. Interactive Map / Spot Listing View:
   The system shall display available study spots in both list view and an interactive campus map view.
4. Crowdsourced Status Updates:
   Registered students shall be able to submit live updates on noise levels and occupancy status for any study spot.
5. Room & Pod Reservation:
   Users shall be able to reserve bookable study rooms/pods for specific time slots with instant confirmation.
6. Admin Spot & Reservation Management:
   Administrators shall be able to create, update, or disable study locations, adjust operating hours, and oversee reservation schedules.

Non-Functional Requirements:

1. Performance:
   Search results and study spot availability queries shall load in under 2 seconds under standard campus traffic.
2. Usability & Accessibility:
   The web interface shall be responsive across mobile, tablet, and desktop devices.
3. Reliability & Availability:
   The system shall maintain an uptime of 99.5% during academic semesters, backed by ASP.NET Core exception logging and database connections.
4. Security:
   All passwords shall be hashed using ASP.NET Core Identity, and user sessions shall use secure HTTPS cookies.

4. Application Design

The application follows a Model-View-Controller (MVC) architectural pattern using ASP.NET Core (C#) for the backend, Entity Framework Core for data access, and SQL Server for storage.



Layer Interaction and Data Flow

1. Front-End Layer:
   Developed using Razor views, HTML, Bootstrap and JavaScript. Renders UI elements for searching spots, viewing maps and managing bookings.

2. Back-End Layer:
   - Controllers handle incoming HTTP requests.
   - Services/Logic Layer enforces business rules (e.g: checking reservation time overlaps).
   - EF Core (Data Access) translates C# LINQ queries into SQL queries executed against the database.

3. DB Layer:
   Microsoft SQL Server stores relational tables for users, spots, buildings, reservations and live updates.

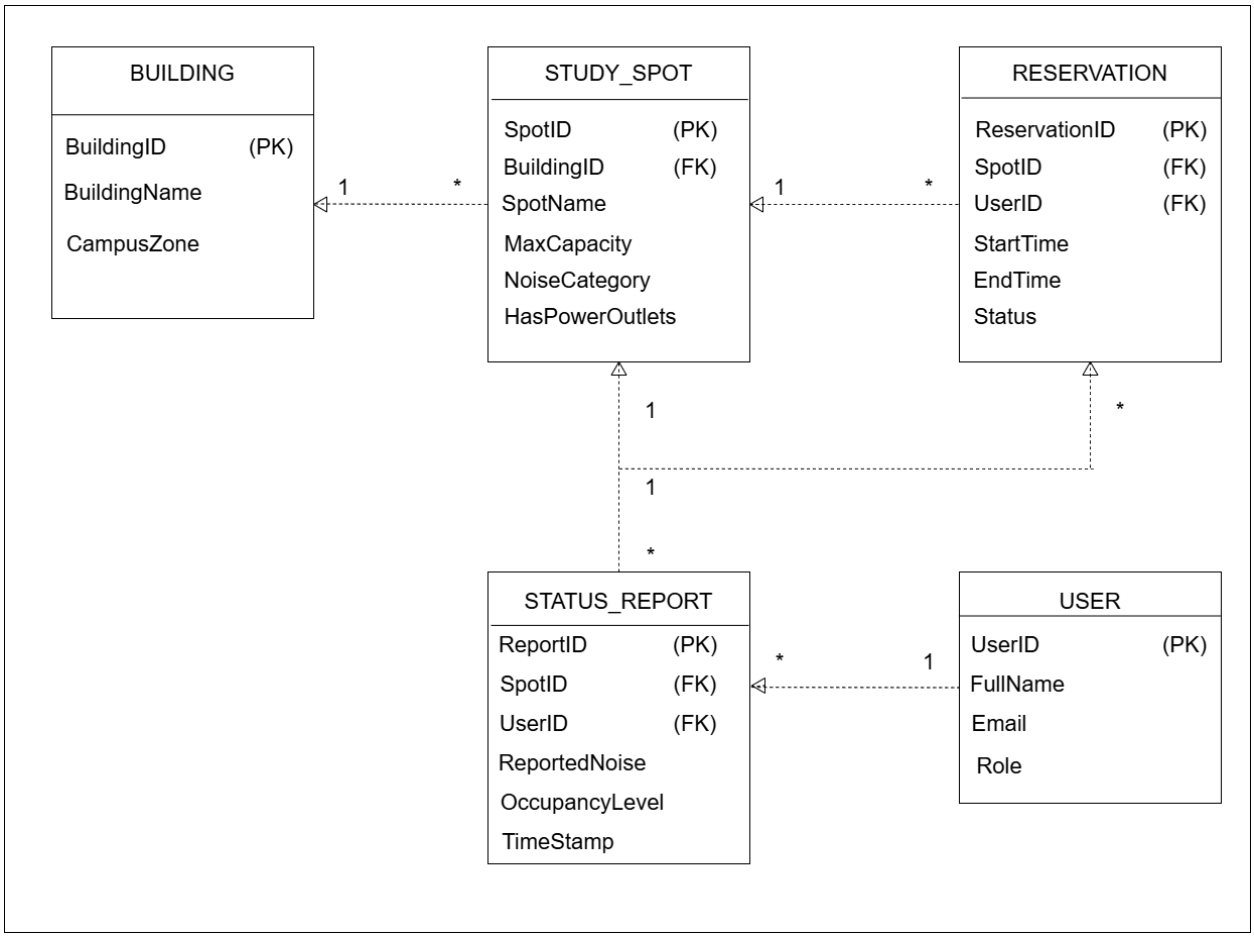
Main Website Pages and Roles

- Home (/Spots):
  Displays the search bar, filter panel (noise, outlets, building) and interactive spot cards with real-time status.
- Spot Detail Page(/Spots/Detail/{id}):
  Shows full details for a spot, including amenities, operational hours, active crowdsourced reports and embedded reservation form.
- Reservation Booking Page (/Reservations/Create):
  Form for Selecting date/time intervals to reserve individual study rooms.
- Admin Dashboard (/Admin/Dashboard):
  Restricted view for admins to add/edit spots, moderate live status reports, and view usage statistics.

5. Database Design

Entity-Relationship (ER) Overview:

- User:
  Holds student and admin account details.
  (one-to-many with StudySpot)
  (one-to-many with StatusReport)
- Building:
  Represents campus buildings
  (one-to-many with Reservations)
  (one-to-many with StatusReport)
- Reservation:
  Tracks booked time slots for study spots users.
- StatusReport:
  Stores real-time crowdsourced reports (noise, occupancy) posted by users.



Database Tables Specification

Table1: Users

Stores account information for students and administrative staff.

Field Name	Data Type	Constraints	Description
UserID	INT	Primary KEy, Auto Increment	Unique user identifier
FullName	NVARCHAR(100)	NOT NULL	Users full name
Email	NVARCHAR(150)	NOT NULL, Unique	University email address
PasswordHash	NVARCHAR(MAX)	NOT NULL	Hashed account password
Role	NVARCHAR(20)	NOT NULL, Default:'Student'	User role('Student', 'Admin')
CreatedAt	DATETIME2	NOT NULL, Default:GETUTCDATE()	Registration timestamp

Table 2: StudySpot

Contains details about specific study locations across campus

Field Name	Data Type	Constraints	Description
SpotID	INT	Primary Key, Auto-Increment	Unique spot identifier
BuildingID	INT	Foreign Key (Buildings.BuildingID), NOT NULL	Associated building
SpotName	NVARCHAR(60)	NULL	E.g: 2nd Floor Silent Corner
RoomNumber	NVARCHAR(20)	NULL	Room/Zone number
MaxCapacity	INT	NOT NULL, Check (MaxCapacity > 0)	Maximum seating capacity
NoiseCategory	NVARCHAR(30)	NOT NULL	‘Quiet’, ‘Moderate’, ‘Loud’
HasPowerOutlets	BIT	NOT NULL, Default: 1	1 = Available, 0 = Unavailable
IsBookable	BIT	NOT NULL, Default: 0	1 = Requires reservation, 0 = Walk-in

Table 3: Reservation

Manages study room/spot bookings made by users

Field Name	Data Type	Constraints	Description
ReservationID	INT	Primary Key, Auto-Increment	Unique reservation identifier
SpotID	INT	Foreign Key (StudySpots.SpotID), NOT NULL	Reserved study spot
UserID	INT	Foreign Key (Users.UserID), NOT NULL	User who made the reservation
StartTime	DATETIME2	NOT NULL	Booking start time
EndTime	DATETIME2	NOT NULL, Check (EndTime > StartTime)	Booking end time
Status	NVARCHAR(20)	NOT NULL, Default: ‘Confirmed’	‘Confirmed’, ‘Cancelled’, ‘Completed’

6. Team Contribution Table

Team Member Name	Student ID	Assigned Tasks and Deliverables	Work Percentage
Ghassan Al-Julanda Al-Mawali	144474	Part A: ● Defined objectives, Motivation and real-world relevance(Q1) ● Conducted user interview 1 and summarized findings(Q2) ● Formed Functional and Non-Functional Requirements(Q3) ● Designed database ER diagram, entities, attributes and relationships(Q5) Part B: ● Defined C# enumeration(Q6) ● Created C# entity classes(Q7) ● Implemented business logic layer and service classes(Q10) ● Configured C# exception handling and error pages(Q12) Part C: ● Setup and managed GitHub repository and permissions(Q17) ● Final Word report and table of contents(Q15)	50%

Al-Mustafa Al-Amri	151920	Part A: ● Conducted user interview 2 and summarized findings(Q2) ● Authored system architecture and layer interaction overview(Q4) Part B: ● Integrated front-end framework with ASP.NET Core(Q8) ● Developed 4 razor web pages(Q9) ● Created custom layout, navigation bar and footer(Q11) ● Applied bootstrap styling, themes and component styling(Q13) ● Configured the form validation(Q14) Part C: ● Managed final submissions on moodle and GitHub(Q17)	50%

GitHub Repository

https://github.com/s144474/StudySpace.git